



Natural Calamity Deaths in Nepal: Analysis of the 2021 National Population and Housing Census

Jibesh Acharya ¹, Srisa Rijal ², and Divya Shakya ³

¹Population and Statistics Research Hub, Nepal

²Pokhara Metropolitan City, Pokhara, Nepal

³Progress Inc.

Received: 10th June, 2026

Accepted: 12th August, 2026

Published: 15th September, 2026

ABSTRACT

Background: Nepal is one of the most disaster-prone countries in South Asia due to its rugged terrain, monsoon-driven precipitation, and active seismicity. However, evidence on the demographic and geographic patterns of mortality associated with natural calamities remains limited. This study examined natural-calamity mortality in Nepal using nationally representative census data, with particular emphasis on differences by age, sex, wealth, residence, province, and district-level geographic exposure.

Method: Data on household-reported natural-calamity deaths were obtained from the 2021 National Population and Housing Census (NPHC). Weighted individual records were used to construct sex-, age-, wealth quintile-, and residence-specific exposure populations. A negative binomial mixed-effects regression model with a log-population offset was fitted to estimate incidence rate ratios, with sex, age group, wealth quintile, place of residence, and province included as fixed-effect predictors. A district-level random intercept was incorporated to account for geographic clustering in natural-hazard exposure.

Results: Among 198,463 deaths recorded in the census, 9,595 (4.83%) were attributed to natural calamities, corresponding to a national mortality rate of 32.9 per 100,000 population. Madhesh Province recorded the highest mortality rate (55.8 per 100,000) and the highest proportion of all deaths attributable to natural calamities (10%). Mortality followed a J-shaped age pattern, with the lowest risk among working-age adults and the highest among individuals aged 75 years and above. Mortality was significantly higher among males than females and among urban than rural residents. Only individuals in the wealthiest households experienced significantly lower mortality than those in the poorest households. No province-level differences remained statistically significant after accounting for geographic clustering below the provincial level. Substantial variation persisted across districts, with the highest-risk districts distributed across several provinces.

Conclusion: Natural-calamity mortality in Nepal is shaped not only by age, sex, wealth, and place of residence but also, importantly, by localized district-level exposure. The absence of significant province-level differences after accounting for district clustering suggests that disaster risk varies considerably within provinces. Disaster-risk-reduction investments and early-warning systems should therefore be targeted at the district level, with evacuation and preparedness strategies tailored to vulnerable age groups and residential settings.

Keywords: natural disasters, mortality, Nepal, census, negative binomial regression

Correspondence: Jibesh Acharya, Population and Statistics Research Hub, Nepal. E-mail: acharyajibesh@gmail.com.

INTRODUCTION

Globally, Nepal is one of the most highly exposed countries to natural hazards. Here, the population is recurrently exposed to natural calamities such as earthquakes, floods, landslides, lightning, glacial lake outburst floods, and others. These events are often triggered by its steep, tectonically active terrain and monsoon-dominated precipitation (Group and Bank., 2021). The frequency of disasters is no less. Between mid-2018 and mid-2024 alone, Nepal recorded more than 32,000 disaster incidents. As a result, 3,672 people died, while many people were missing, and multiple communities were displaced (of Home Affairs., 2024). Single catastrophic events can also produce mortality shocks that far exceed the usual and expected annual burden. The recent and widely known historical event is the 2015 earthquake, which killed an estimated 8,951 people across 41 districts (Ton-

nelier and Charalampous, 2026). That earthquake also left a lasting impact of secondary hazard exposure. It destabilized hill slopes that remain landslide-prone today (Adhikari and Mishra, 2021). More recent events have followed the same pattern. The November 2023 Jajarkot earthquake killed 153 people in Karnali Province (Gautam, 2023), the late-September 2024 floods and landslides killed more than 215 people across 44 districts (Nations., 2024), a supraglacial lake outburst on the Bhote Koshi in July 2025 left at least nine people dead and more than two dozen missing in Rasuwa (Reuters., 2025), and unseasonal October 2025 rainfall killed at least 60 people in eastern Nepal, 39 of them in Ilam district alone (Gautam, 2025). Recurrent hazards add a further, less visible burden. Lightning alone kills of the order of 100 people every year and ranks among the deadliest hazards in the country (Adhikari, 2021; Sharma and Cooper,

2022). The Sendai Framework for Disaster Risk Reduction 2015–2030 targets a substantial reduction in global disaster mortality. Disaster-attributable death rates are observed under Sustainable Development Goal indicators 1.5.1, 11.5.1, and 13.1.1. This reflects international recognition of natural disasters as a measurable and preventable contributor to premature mortality (for Disaster Risk Reduction., 2015).

Existing evidence on disaster-related mortality in Nepal comes mostly from event-specific and hazard-specific studies. These include retrospective analyses of an event and rarely incorporate all hazard types together in a single estimate (Chapagain and Borgemeister, 2024; Gaire and Arcos González, 2015; K.C., 2013). Together, a consistent set of structural vulnerabilities is identified. These include settlement on river banks, particularly in flood plains, unstable hill slopes, and limited early-warning and evacuation infrastructure in vulnerable regions. They also highlight gender differentials in disaster mortality. Compared to men, women in South Asia are estimated to have a higher disaster-related mortality rate. It is attributed to mobility restrictions, caregiving responsibilities, and lower access to early-warning information (Neumayer and Plümper, 2007; for Asia and the Pacific., 2022). Besides deaths occurring inside health facilities, the cause of death is not routinely recorded in Nepal. This limits what routine systems can report (Office., 2023). As a result, geographically disaggregated estimates of natural calamity deaths across the country remain limited. How demographic and socioeconomic characteristics along with geography jointly shape this risk is also not well quantified.

The Nepal Population and Housing Census (NPHC) 2021 addressed part of this gap. At national scale, it asked households to report whether a death occurring in the preceding 12 months was due to a natural calamity. Household-reported cause of death carries well-known limitations relative to medically certified or event-verified causes. However, it provides population-level data on natural calamity deaths for the whole country. These data span whichever hazards happened to affect a household in the reference period. Thus, this study uses these data to (i) understand the geographical distribution of natural calamity deaths in Nepal in 2021, and (ii) identify the socio-demographic predictors of those deaths and quantify the risk associated with them.

METHODOLOGY

Study Design, Setting, and Data Sources

This is a cross-sectional study using the sample data from the National Population and Housing Census (NPHC) 2021 (Office., 2023). The census enumerated all usual residents of Nepal. It also recorded detailed information on the deaths of household members occurring during the 12 months preceding the census reference date (25 November 2021).

The unit of analysis was a demographic stratum. It was created by cross-classifying individuals by sex, age group, place of residence, and household wealth quintile. For each stratum, the number of deaths due to natu-

ral calamities and the corresponding weighted exposure population were computed from the deceased and individual census files, respectively. The two files were then merged by their common stratifying variables. A few sparsely populated strata contained a reported death but no matching exposure population, because the weighted individual file held no record with that exact combination of characteristics. Dropping these strata would have discarded observed deaths, so each was assigned a nominal population of one. This affected a small number of strata and did not significantly change the model estimates.

Outcome Variable

The outcome variable was the number of deaths due to natural calamities recorded per stratum. During the census, a respondent was asked to classify each household death that occurred in the preceding 12 months into one of the following categories: communicable disease, non-communicable disease, road traffic accident, other accident, pregnancy-related, crime, suicide, natural calamities, or other. For this study, deaths reported as due to natural calamities were taken as the outcome, and deaths from all remaining reported categories were treated as the comparison group. The “natural calamities” category is hazard-agnostic. It may include deaths from floods, landslides, earthquakes, avalanches, lightning, and other acute environmental hazards, without further subclassification by hazard type. Cause of death was reported by a household member rather than medically certified or verified against a disaster registry. The outcome should therefore be read as household-attributed rather than officially verified disaster mortality.

Predictor Variables

Five predictor variables were included in the analysis. They were:

Age (0–4, 5–14, 15–24, 25–34, 35–44, 45–54, 55–64, 65–74, 75 years and above)

Sex (Male and Female)

Place of residence (Rural, Urban, Peri-urban)

Household wealth quintile (Lowest, Lower, Middle, Higher, Highest)

Province (Bagmati, Gandaki, Karnali, Koshi, Lumbini, Madhesh, Sudurpaschim)

Exposure populations for each stratum were derived from weighted individual-level census records.

Statistical Analysis

Firstly, national, provincial, and district-level natural calamities deaths were summarized using counts, crude death rates per 100,000 population, and the proportional share of natural calamities among all recorded deaths. A crude rate is the ratio of the observed number of deaths and the exposed population. District rates and proportions were mapped using shapefiles for administrative boundaries current at the time of the 2021 census.

To estimate the association between socio-demographic characteristics and natural calamity mortality, a negative binomial mixed-effects regression with a log link was fitted:

$$\log(E[Y_i]) = \log(\text{Population}_i) + X_i\beta + u_{d(i)}$$

where,

Y_i is the number of natural-calamity deaths in stratum i ,

$Population_i$ is the corresponding enumerated population entered as a log-offset, and

X_i is the vector of fixed-effect predictors (sex, age group, household wealth quintile, place of residence, and province).

$u_{d(i)} \sim N(0, \sigma_d^2)$ is a random intercept for the district d containing stratum i .

Province was included as a fixed effect to assess geographic variation in disaster risk directly. A district-level random intercept was added because natural-calamities mortality is expected to cluster geographically at a finer scale than a large provincial catchment. Models were fitted with the negative binomial (type 2) family, specified as `nbinom2` in the R package `glmmTMB` (Brooks and Bolker, 2017). This family was chosen over a standard Poisson specification because of overdispersion, since the counts showed substantially greater variance than the mean. Overdispersion is expected for disaster deaths, which concentrate in a small number of severely affected strata rather than spreading evenly across the population.

Model Evaluation and Diagnostics

Multicollinearity among predictors was assessed using generalized variance inflation factors (GVIFs) (Fox and Monette, 1992). All adjusted GVIF values fell below the conventional threshold of 2, indicating no problematic collinearity (Craney and Surles, 2002). Incidence rate ratios (IRRs) with 95% confidence intervals were obtained by exponentiating fixed-effect coefficients.

A preliminary fixed-effects-only specification (sex, age group, wealth quintile, residence, and province, without a district term, fitted on province-aggregated strata) showed a Pearson dispersion ratio of 7.50. This shows that the negative binomial family alone did not adequately capture stratum-to-stratum clustering. A district-level random intercept reduced this ratio to 1.94 and is retained in the model reported throughout the study. Model adequacy was assessed using the marginal and conditional for generalized linear mixed models, computed with the R package `performance`. The marginal provides variance explained by the fixed effects alone, and the conditional adds the district random intercept. All analyses were conducted in R version 4.5.0 (2025-04-11 ucrt) using `glmmTMB` 1.1.14, `performance`, and `broom.mixed` packages.

RESULTS

Geographic Distribution of Natural Calamities Mortality

National and provincial variation

Provincial variation was marked. Madhesh recorded the highest crude death rate (55.8 deaths per 100,000 population), followed by Lumbini (35) and Gandaki (32.4). Karnali had the lowest rate (22.2 deaths per 100,000 population) (Table 1). Madhesh also carried the highest proportional share of all deaths due to natural calamities

Table 1. National and provincial natural calamities mortality, Nepal, 2021.

Region	Total deaths	Number of deaths	Death rate (per 100,000 population)	Proportion of deaths (%)
Nepal (national)	198,463	9,595	32.9	4.8
Madhesh	34,319	3,415	55.8	10
Lumbini	36,207	1,791	35	4.9
Gandaki	20,406	798	32.4	3.9
Koshi	38,275	1,215	24.5	3.2
Sudur paschim	16,988	640	23.7	3.8
Bagmati	43,872	1,362	22.3	3.1
Karnali	8,396	374	22.2	4.5

ties (10%), more than double that of any other province. Both measures were therefore high in the same province.

District-level variation

Natural-calamities mortality rates ranged from 13.3 to 68.5 deaths per 100,000 population across districts (Fig. 1). Mahottari had the highest rate (68.5 per 100,000 population), followed by Saptari (66.7), Gorkha (60.6), and Sarlahi (59). Most of the highest-rate districts lie in Madhesh Province. Gorkha is an exception, because it is a hill district rather than a Terai one. Humla, a sparsely populated mountain district, recorded the highest proportional share of deaths due to natural calamities (13.4%), despite a modest rate.

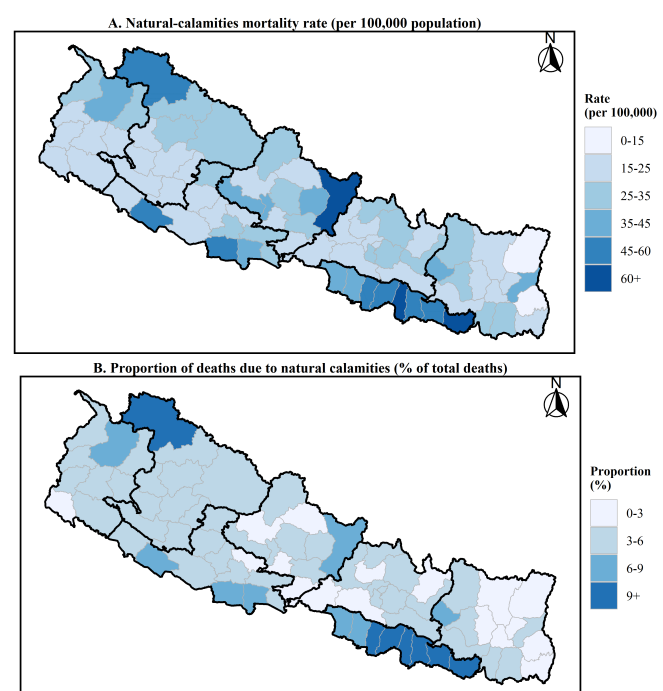


Figure 1. Natural-calamities mortality by district, Nepal, 2021. Provincial boundaries are overlaid in black.

Table 2. Descriptive summary of socio-demographic characteristics of the study population, Nepal census 2021

Variable	Category	Exposure population	Natural calamities deaths	Rate (per 100,000)	Share of deaths (%)
Sex	Female	14,919,849	4,009	26.87	4.8
	Male	14,244,729	5,578	39.16	4.9
Age Group	0–4	1,909,507	575	30.11	5.7
	5–14	5,619,326	343	6.10	8.5
	15–24	5,838,337	321	5.50	4.4
	25–34	4,767,417	328	6.87	4.1
	35–44	3,893,350	491	12.62	3.8
	45–54	2,873,104	474	16.50	2.4
	55–64	2,109,330	832	39.45	3.0
	65–74	1,438,654	1,777	123.51	4.3
	75+	715,553	4,446	621.33	6.6
Wealth Quintile	Lowest	5,757,063	2,369	41.15	6.2
	Lower	5,918,418	1,887	31.88	4.8
	Middle	5,991,255	2,211	36.90	5.4
	Higher	5,465,884	1,575	28.81	4.5
	Highest	5,785,401	1,545	26.71	3.5
Place of Residence	Rural	9,642,797	2,614	27.10	3.9
	Urban	7,964,905	2,194	27.55	4.1
	Peri-urban	11,556,877	4,779	41.35	6.2

* The natural-calamities deaths shown here total 9,587, which is 8 fewer than the 9,595 recorded nationally because deaths for which age was not stated were excluded from the analysis.

Descriptive Characteristics of the Study Population

Table 2 presents the distribution of natural calamity mortality by sex, age group, household wealth quintile, and place of residence. The mortality rate was higher among males than females (39.16 vs 26.87 per 100,000 population), but had almost similar proportions of deaths due to natural calamity. The mortality was high among the youngest children, declining to its lowest level among those aged 15–24 years (5.50 per 100,000), and increasing steadily thereafter. The rate reached 621.33 per 100,000 among adults aged 75 years and above. This is more than 100 times the rate observed among those aged 15–24 years. Although children aged 5–14 years had a low mortality rate (6.10 per 100,000), natural calamities accounted for the largest share of all deaths in this age group (8.5%). This relatively high share reflects the low underlying mortality at these ages. As a result, a small number of natural calamity deaths reflects a comparatively large proportion of total deaths.

A socioeconomic gradient was also evident. Generally, mortality was higher among individuals from low household wealth quintiles. The highest mortality rate was observed in the lowest wealth quintile (41.15 per 100,000), compared with 26.71 per 100,000 in the highest quintile. Mortality also differed by place of residence. Peri-urban areas had the highest rate (41.35 per 100,000), substantially exceeding the rates in rural (27.10) and urban (27.55) areas. Overall, a marked difference in natural-calamity mortality across socio-demographic variables, with particularly elevated rates among older adults, males, poorer populations, and those living in peri-urban areas, was observed.

Table 3. Generalized variance inflation factors for predictors included in the regression model.

Predictors	GVIF	DF	Adjusted GVIF
Sex	1.00	1	1.00
Age Group	1.05	8	1.00
Wealth Quintile	1.94	4	1.09
Place of residence	2.66	2	1.28
Province	2.21	6	1.07

Predictors of Natural-Calamities Mortality

Multicollinearity diagnostics indicated no evidence of problematic collinearity. All adjusted GVIF values fell below the conventional threshold across model covariates (Table 3)

Predictors of Natural-Calamities Mortality

Table 4 presents adjusted incidence rate ratios from the final negative binomial mixed-effects regression model. Age showed a J-shaped association with mortality. Relative to children aged 0–4, risk fell sharply through adolescence and working age. The population aged 15–24 years had the lowest risk (IRR = 0.27, 95% CI: 0.14–0.50). The risk was high for the population older than 55 years, with the highest among those aged 75 years and above (IRR = 13.33, 95% CI: 7.89–22.51). This oldest group faced the highest adjusted mortality of any age group. Females had significantly lower mortality risk than males after adjustment (IRR = 0.70, 95% CI: 0.55–0.90). Household wealth showed a protective effect among the highest group. Only the highest quintile differed significantly from the lowest (IRR = 0.57, 95% CI: 0.36–0.91). Place of residence was strongly associated with mortality. Urban residents had

Table 4. Adjusted incidence rate ratios (IRR) for natural-calamities mortality from the negative binomial regression model

Variable	Category	Adjusted IRR (95% C.I.)
Sex	Female	0.70 (0.55–0.90)**
Age Group	5–14	1.18 (0.59–2.39)
	15–24	0.27 (0.14–0.50)***
	25–34	0.37 (0.20–0.70)**
	35–44	0.33 (0.18–0.60)***
	45–54	0.42 (0.23–0.74)**
	55–64	1.20 (0.68–2.11)
	65–74	2.27 (1.34–3.84)**
Wealth Quintile	75+	13.33 (7.89–22.51)***
	Lower	0.82 (0.57–1.18)
	Middle	0.69 (0.47–1.00)
	Higher	0.76 (0.51–1.15)
	Highest	0.57 (0.36–0.91)*
Place of Residence	Urban	2.57 (1.84–3.60)***
	Peri-urban	1.26 (0.89–1.76)
Province	Bagmati	0.91 (0.39–2.10)
	Gandaki	0.90 (0.37–2.19)
	Karnali	0.92 (0.35–2.41)
	Koshi	0.55 (0.24–1.29)
	Lumbini	0.59 (0.25–1.37)
	Sudurpaschim	0.45 (0.17–1.14)

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Reference categories: Sex = Male; Age = 0–4 years; Wealth Quintile = Lowest; Place of Residence = Rural; Province = Madhesh. The model includes a log-population offset.

substantially higher mortality risks than rural residents (IRR = 2.57, 95% CI: 1.84–3.60). Peri-urban residents did not differ significantly from rural residents after adjustment (IRR = 1.26, 95% CI: 0.89–1.76). Lastly, none of the provinces differed significantly from Madhesh regarding mortality.

Table 4: Adjusted incidence rate ratios (IRR) for natural-calamities mortality from the negative binomial regression model.

Model Diagnostics

Table 5. Districts with the highest and lowest modelled excess risk of natural-calamities mortality

Rank	District	Province	Excess risk (IRR)
Highest	Rasuwa	Bagmati	7.69
Highest	Taplejung	Koshi	4.74
Highest	Lamjung	Gandaki	3.45
Highest	Humla	Karnali	3.08
Highest	Bajhang	Sudurpaschim	2.91
Lowest	Tanahu	Gandaki	0.42
Lowest	Sankhuwasabha	Koshi	0.40
Lowest	Jhapa	Koshi	0.36
Lowest	Nawalparasi East	Gandaki	0.22
Lowest	Surkhet	Karnali	0.19

The district-level random intercept had a standard deviation of 0.83 on the log-count scale, indicating substantial district-to-district variation in mortality that is not explained by demographic composition or province membership. Table 5 presents the five districts each with the highest and the lowest modelled excess risk. Rasuwa (Bagmati province, IRR = 7.69) carried the largest excess risk, and Surkhet (Karnali, IRR = 0.19) the smallest. The five highest-risk districts span five different provinces, which shows that the strongest geographic signal in these data operates at the district level and does not align cleanly with provincial boundaries. Full estimates for all 77 districts are given in Annex 2.

Table 5: Districts with the highest and lowest modelled excess risk of natural-calamities mortality

DISCUSSION

This study used household-reported cause of death from the 2021 census to assess variation in natural calamity mortality across Nepal. Three key findings were found. First, natural calamity mortality varied strongly by age, with the highest risk among the elderly population. Second, males had higher mortality than females, while urban residents had higher mortality than rural residents after adjustment. Third, provincial differences were not statistically significant after accounting for geographic clustering, although mortality varied substantially across districts.

Age was the strongest pattern in the analysis. Compared with children aged 0–4 years, mortality was lower among people aged 15–54 years. It increased again among older adults and was more than 13 times higher among those aged 75 years and above. This pattern is consistent with greater vulnerability among very young and older populations. Young children and older adults may have greater difficulty escaping rapidly developing hazards and may depend more on others during evacuation (Tonnelier and Charalampous, 2026). Studies of floods, landslides, and the 2015 earthquake in Nepal have also reported greater vulnerability among older populations (Adhikari and Mishra, 2021; Gaire and Arcos González, 2015).

Mortality was lower among females than males after adjustment. This finding differs from studies in South Asia that report higher female mortality during disasters (Neumayer and Plümper, 2007; for Asia and the Pacific., 2022). Gender differences have also been examined in Nepal, in years-of-life-lost analyses of the 2015 earthquake (Tonnelier and Charalampous, 2024), though that evidence covers a single seismic event rather than the recurrent floods and landslides that drive most of the deaths recorded here. One possible explanation is the hazard profile in Nepal. Floods and landslides are common, recurrent hazards in Nepal (Gaie and Arcos González, 2015; Pokharel and Raule, 2026), and men may have greater exposure through outdoor and agricultural work. Evidence from Nepal also shows a higher concentration of lightning deaths among men, particularly those engaged in outdoor activities (Adhikari, 2021; Sharma and Cooper, 2022). However, the census does

not record the hazard type, location of death, or activity at the time of death. These scenarios therefore cannot be tested in the present study.

The association with household wealth was weaker after adjustment than in the descriptive analysis. Only the highest wealth quintile had significantly lower mortality than the lowest quintile. Poorer households may have greater exposure to hazardous locations and fewer resources for safe housing, evacuation, and recovery (K.C., 2013).

Urban residents had higher adjusted mortality than rural residents. This contrasts with the similar crude mortality rates observed across residence categories. The difference may partly reflect the age and socioeconomic composition of urban populations. Rapid urban growth may also increase exposure to flooding in densely settled and poorly drained areas. Recent flooding in the Kathmandu Valley illustrates this concern (Lamichhane, 2025).

Madhesh had the highest provincial mortality rate and the highest adjusted point estimate. However, none of the provincial estimates differed significantly from Madhesh after accounting for geographic clustering. The apparent provincial differences in a model without district-level clustering were therefore not robust. This indicates that geographic variation in natural-calamity mortality is better captured at a finer spatial scale than the province.

The district-level results support this interpretation. High-risk districts were found across several provinces and were not concentrated in Madhesh. Some districts with high crude rates were located in the hills, while modelled district effects also identified excess risk in hill and mountain areas. These patterns suggest that natural-calamity mortality is not confined to the Terai floodplain. Instead, risk varies across different geographic settings with different hazard profiles. Flooding is a major concern in the Terai, while landslides and other terrain-related hazards are important in hill and mountain districts (Group and Bank., 2021; Gaire and Arcos González, 2015).

The district estimates should nevertheless be interpreted with caution. The census records deaths reported by households but does not identify where the death occurred or the specific hazard involved. The 12-month reference period also prevents the observed district patterns from being linked to individual disaster events. Events occurring during or close to the reference period, such as the 2021 Melamchi flood and the October 2021 rainfall events, may have contributed to some of the observed variation (Chen and West, 2025; Press., 2021; Bank., 2022). These findings therefore indicate where mortality was concentrated rather than which hazards caused the excess risk.

These findings have implications for disaster-risk-reduction and public health planning in Nepal. Preparedness should give particular attention to young children and older adults, who showed the greatest mortality risk. Urban areas also require greater attention to flood-resilient infrastructure and land-use planning. Geographic targeting should consider districts rather than

relying only on provincial averages because high-risk areas were distributed across several provinces. Hazard-specific information should be incorporated into district-level planning where available.

Strengths and Limitations

The main strength of this study is its use of population-level census data covering the whole country, analysed with a model that adjusts jointly for individual, household, and geographic characteristics, including a district-level random intercept for geographic clustering below the province level. It captures natural-calamity deaths across all provinces and hazard types within a common data source, and allows mortality to be examined across several socio-demographic characteristics.

Several limitations should be considered. First, causes of death were reported by household members rather than medically certified or independently verified. Some deaths may therefore have been misclassified. Second, the census does not identify the specific hazard responsible for each death. Floods, landslides, earthquakes, and other hazards have different risk profiles, but they are combined in the outcome. Third, the analysis is cross-sectional. The observed associations should not be interpreted as individual-level causal effects. The available variables also limit the analysis: housing conditions, proximity to rivers or unstable slopes, evacuation access, and occupation at the time of death were not available. Fourth, the district random intercept absorbs geographic clustering statistically but does not explain it: it does not identify which hazard, or which local factor, drives a given district's excess risk. Districts with small populations record few deaths and may have unstable rates, both in the descriptive summaries and in the corresponding random-effect estimates.

CONCLUSION

This study used census data to measure natural-calamities deaths across Nepal. Risk was concentrated at the extremes of the age distribution, among young children and the elderly population. Compared to provinces, mortality varied markedly by district. Males died from natural calamities more often than females. Urban residents faced significantly higher risk than rural residents. These patterns point to three priorities. Disaster-risk-reduction investment should target specific high-risk districts, informed by hazard-specific data, rather than being allocated at a higher administrative tier. Evacuation planning should account for households with young children or older residents. Rapidly urbanizing municipalities face growing, not diminishing, flood exposure and should be planned for accordingly. Stronger cause-of-death reporting would refine these estimates further. Linking future census or survey rounds with Nepal's disaster event databases would let deaths be verified against recorded events and disaggregated by hazard type, tracing the district-level risk concentration identified here to specific hazards. This would support ongoing monitoring in line with the Sendai Framework's mortality-reduction targets.

DECLARATIONS

Ethics Approval and Consent to Participate

This study used publicly available, anonymised census data. No individual-level identifiers were retained in the analytical dataset. Formal ethical review was not required under applicable national guidelines for secondary analysis of anonymised population data.

Consent for Publication

Not applicable.

Availability of Data and Materials

The study used secondary census microdata and publicly available district-level indicators. Access to the census microdata is governed by the National Statistics Office of Nepal. Derived analytic code and inputs are available from the corresponding author on reasonable request.

Competing Interests

The authors declare that they have no competing interests.

Funding

No specific funding was reported for this study.

Authors' Contributions

JA conceptualized the study, curated the data, conducted the statistical analysis, and drafted the manuscript. SR and DS contributed to data interpretation and critical revision of the manuscript. All authors read and approved the final manuscript.

ACKNOWLEDGEMENTS

The authors acknowledge the National Statistics Office of Nepal for conducting the National Population and Housing Census 2021 and making census resources available for research use. The authors also thank Prof. Dr. Samir K.C. for reviewing the manuscript and providing helpful feedback. Claude (Anthropic; Opus 5 and Sonnet 5) was used as an auditable coding, drafting, and revision-support tool during manuscript preparation, including statistical model specification and diagnostic testing. Authors reviewed and take responsibility for all content.

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Citation: Acharya, J., Rijal, S., & Shakya, D.(2026). Natural Calamity Deaths in Nepal: Analysis of the 2021 National Population and Housing Census. *Nepal Journal of Statistics and Data Sciences*. 1(1),57–64. DOI: [10.68031/njsd.v1i1.00007](https://doi.org/10.68031/njsd.v1i1.00007)